Rathina Vel Balasubramanian

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WORK PERMIT: RWR+ (not bound to employer) available to work from October 2026

PROFESSIONAL SUMMARY

Bioengineer with 5 years of hands-on experience in 3D tissue models, hydrogels and scaffolds, working with primary human cells, disease models and 384-well workflows. Motivated to build scalable, reproducible human tissue systems and high-throughput organoid models that replace animal testing in human-first drug discovery.

TECHNICAL SKILLS

3D Tissue Models: Primary human cells (MSCs, Keratinocyte, Fibroblast), spheroids, 3D tissue constructs, disease modelling, co-culture systems, organoid engineering, inflammatory disease models, immunology
Biomaterials and Fabrication: Hydrogel, scaffold design, two-photon polymerization (2PP), 3D bioprinting, electrospinning, CAD (AutoCAD)
Assays and Automation: Cell-based assay development, multi-well plate formats (384-well), automated media handling, high-throughput screening, automation, plate architecture, perfusion
Analytics and Software: Confocal imaging, flow cytometry, histology, RT-qPCR, Western blot, rheology, mechanical testing, GraphPad Prism, ImageJ/Fiji
Languages: English (C2), Tamil (native), Hindi (B1), Malayalam (A2), German (A1, learning)

EDUCATION

PhD in Natural Sciences - *TU Wien, Austria* *07/2023 - 10/2026*
MSc in Biotechnology - *Vellore Institute of Technology, India* *06/2018 - 11/2020*
BSc in Microbiology - *Madurai Kamaraj University, India* *05/2015 - 06/2018*

WORK HISTORY

Doctoral Researcher 07/2023 - 10/2026
TU Wien, 3D Printing and Biofabrication Group *Vienna, Austria*

- Developed injectable scaffolded spheroids from MSCs of 5 donors across 384-well plate formats for treating intervertebral disc (IVD) degeneration.
- Designed an *ex vivo* degenerative disc disease model with the AO Research Institute Davos to validate the therapy based on scaffolded spheroids.
- Operated and maintained high throughput custom 2PP 3D printers and designed 384-well plate architectures for automated sorting workflows.
- Wrote lab SOPs and trained 2 students on fabrication, culture and assay workflows

Project Associate 07/2021 - 05/2023
CSIR - Central Glass and Ceramics Research Institute *Kolkata, India*

- Engineered electrospun nanofibrous scaffolds with sugar-glass nanoparticles to deliver growth factors for cartilage regeneration; co-invented an international patent on a composite cartilage scaffold

Project Scientist 03/2021 – 07/2021
Sree Chitra Tirunal Institute of Medical Sciences and Technology *Thiruvananthapuram, India*

- Identified the bioink component that improved shear-thinning and print accuracy by screening six composite hydrogels for extrusion bioprinting; published as first author in Materials Today Communications

PROJECTS

MSC Spheroids for IVD-like Differentiation - *2PP, MSC spheroids, RT-qPCR, Histology* 06/2023 - 01/2025

- Optimized MSCs spheroids with 2PP microscaffolds for IVD-like differentiation

GDF5-Functionalised Microscaffolds - *2PP, HA chemistry, RT-qPCR, Histology* 01/2025 – 07/2026

- Functionalised microscaffolds with hyaluronic acid to present GDF5 and drive IVD-like differentiation of MSC spheroids

Vascularised Disc-on-Chip Model - *2PP, AutoCAD, disease modelling* 2026

- Conceived a multi-organoid disc-on-chip model integrating 2PP-printed vessel structures and automated media handling workflows to screen disease treatments.

PUBLICATIONS & PATENTS

8 peer-reviewed papers
Patent WO PCT/IN2025/050650 (composite cartilage scaffold)

REFERENCES

Name: Prof. Aleksandr Ovsianikov
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